

• encoding Categorical Features

• Label encoding: Ordinal, binary Categories like t-shirt sizes.

• One-hot encoding: do not have any order to them

it would create one column for each categories

Sklearn. Pre processing → Label Encoder, One Hot encoder

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۱	۲	۳	۴	۵	۶	۷	۸	۹	۱۰	۱۱	۱۲	۱۳	۱۴
۱۵	۱۶	۱۷	۱۸	۱۹	۲۰	۲۱	۲۲	۲۳	۲۴	۲۵	۲۶	۲۷	۲۸
۲۹	۳۰	۳۱											



Correlation Coefficients : r

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum (x_i - \bar{x})^2)(\sum (y_i - \bar{y})^2)}}$$

how do we find out what type of categorical data that we have?

• first we look in to the data and we use unique() method

if there are Binary then just use LabelEncoder.

Binary / multi Class

binary → Label Encoder

multi Class $\begin{cases} \text{Ordinal} \rightarrow \text{Label Encoder} \\ \text{nominal} \rightarrow \text{One-hot Encoding} \end{cases}$

if in a dataset and inside your target

Variable you have Categories you should

encode them based on what we learned.

mo	tu	we	th	fr	sa	su	mo	tu	we	th	fr	sa	su
		1	2	3	4	5	6	7	8				
9	10	11	12	13	14	15	16	17	18	19	20	21	22
23	24	25	26	27	28	29	30	31					

fit & LabelEncoder, X-train فقط
 y-train, X-train & own fit بعد اون
 : fit & transform

le = LabelEncoder()

le.fit(X_train)

X_train_encoded = le.transform(X_train)

X_test_encoded = le.transform(X_test)

train data → median age = 50

test data → median age = 70

You should not Replace test data
 age with median of 70, it should

be Replaced with train data median

50!! NOT 50 in train

& 70 in test

50 in Both places!

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The Encoders will be Saved for Later Predictions on new data.

ONE HOT ENC → Like Colors
→ Embarked cities etc.

drop = "first"

↳ Dummy Variable drop.

e.g

Color column

{ red → Color-red
blue → Color-blue
green → Color-green

red

↳ red

blue

green

Color-red

1

0

0

Color-blue

0

1

0

Color-green

0

0

1

رادرور زرتشت پیامبر

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⇒ when you think about it one of columns are redundant if two variables are 0 then it is the first one

3 categories → two dummy columns (Variables)

n categories → n-1 dummy variables.

↳ multicollinearity.

mo	tu	we	th	fr	sa	su	mo	tu	we	th	fr	sa	su
					1	2	3	4	5	6	7	8	
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Gender → nominal category
→ it is binary!

because we used LabelEncoder and it is Nominal then we get 1 dummy Column, it is same as ~~LabelEncoder~~.

Label Encoder

handle_unknown=False → imagine in your train data you have green, blue, red and one instance of color Black is in test set and not in train set, you can prevent from getting an error with this parameter in place. it would put all values as zero

sparse_output = False.

it get all 1's and 0's. Not just 1!

Color :

red

Color-red

Color-green Color-Blue

Blue

⇒

this would be deleted

Green

it would increase the df dimensionality!